Statistics 280 — Practice Midterm 1B

Unofficial practice exam, written in the style of the W25 and W26 Midterm 1.
Covers Part 1: study design, descriptive statistics, the normal curve, Markov & Chebyshev, correlation, regression.

Name _____ Start time / End time _____

- Answer all questions in the space provided and indicate where you have written additional work.
- Solutions without justification will be marked as wrong for all questions except the scatter-plot matching question. Show all work. A calculator command does not count as justification.
- Express answers as fractions or as decimals rounded to 2 decimal places.
- The last page is a table of values from the normal distribution. Closed book, no notes. A calculator that can take square roots is allowed.
- Suggested time: 80 minutes. Total: 200 points.

Question	**1**	**2**	**3**	**4**	**5**	**6**	**Total**
Points	30	30	35	35	30	40	200
Score							

1. (30 points) A state awards a full-tuition college scholarship to every high-school senior who scores 1200 or above on a statewide exam. Researchers want to know whether the scholarship increases the chance of graduating from college within six years.

Exam score	Students	Graduated within 6 years
1200 or above (scholarship)	12,000	72%
Below 1200 (no scholarship)	48,000	41%

a) (7 points) Can we conclude that the scholarship raises the graduation rate by 31 percentage points? Name two likely confounders and explain why each one qualifies as a confounder.

b) (8 points) The researchers now look only at students who scored 1180–1199 (1,500 students, 58% graduated) and 1200–1219 (1,400 students, 63% graduated). What is this study design called? Why is this comparison more credible, what is the estimated effect, and what should the researchers check before trusting it?

c) (7 points) Suppose students who scored 1190–1199 were allowed to retake the exam, and many of the most motivated ones did, pushing their scores above 1200. How would this affect the comparison in part b)? In which direction would it bias the estimate?

d) (8 points) A different team uses cross tabulation on all 60,000 students, controlling for family income (4 categories), sex (2 categories), urban/rural (2 categories), and school type (public, private, charter). They want every comparison to include at least 20 scholarship recipients and 20 non-recipients. What is the minimum total sample size? Even with plenty of data, what problem with cross tabulation remains?

2. (30 points)

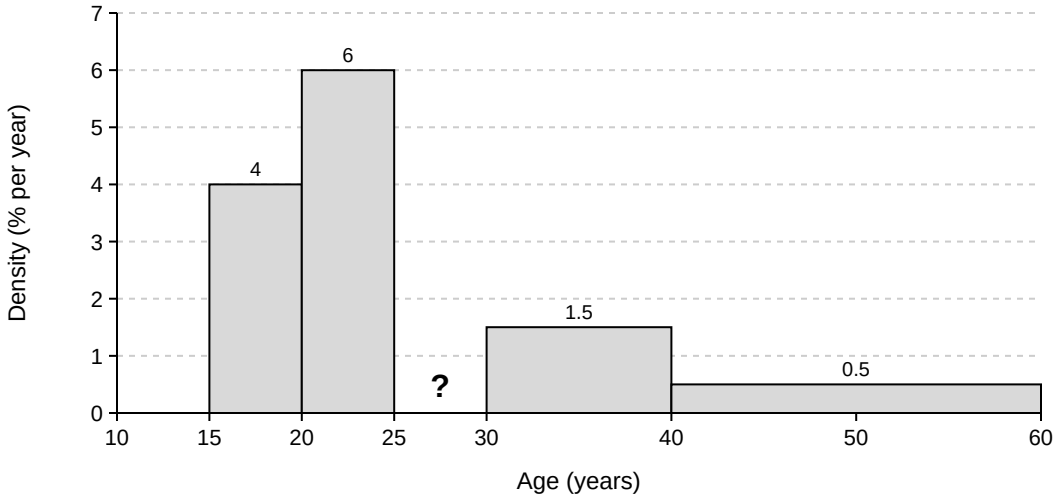
a) (6 points) A course has three sections with 20, 30, and 50 students. The section averages on a midterm are 80, 70, and 76. Find the average for the whole course. A TA instead averages the three section averages. What does she get, and why is it wrong?

b) (8 points) The SDs within the three sections are 5, 10, and 6. Find the SD of all 100 scores. Why is it larger than the SD of any single section? (Hint: shortcut formula.)

c) (10 points) Consider the list 12, 3, 7, 7, 20, 1, 9, 15, 4, 7. Using the definition of percentile from class, find the 25th percentile, the median, the 75th percentile, the IQR, the 90th percentile, the 91st percentile, and the MAD (median absolute deviation from the median).

d) (6 points) Find the number a such that, when a is added to the list in part c), the mean of the new list equals its median.

3. (35 points) The histogram below shows the ages of the 400 people at a concert. The block for ages 25 to 30 has been erased. The numbers above the blocks are heights in percent per year. Ages are evenly distributed within each block, and each interval includes its left endpoint but not its right.



a) (5 points) Find the height of the missing block. How many people are 25 to 30 years old?

b) (6 points) Estimate the number of people between 22 and 35 years old.

c) (8 points) Estimate the median and the mean age.

d) (8 points) Estimate the SD of the ages by treating every person in a block as if they were at the block's midpoint. Is the true SD likely to be larger or smaller than your estimate? Explain.

e) (8 points) R draws a Tukey boxplot: whiskers extend to the most extreme values within 1.5 IQR of the quartiles, and values beyond that are plotted as outliers. Estimate the quartiles. Would a 55-year-old be plotted as an outlier? Estimate the percentage of attendees plotted as outliers. Does the lower whisker reach the youngest attendee?

4. (35 points) Battery lifetimes follow the normal curve with mean 500 hours and SD 40 hours (parts a–d).

a) (5 points) Estimate the percentage of batteries that last between 450 and 560 hours.

b) (5 points) The manufacturer will replace any battery that fails before T hours. Find T so that only 2.5% of batteries are replaced.

c) (5 points) A battery lasts 530 hours. Approximately what percentile is it?

d) (6 points) Engineers improve the process. The mean stays at 500 hours, but now only 5% of batteries last less than 450 hours. Estimate the new SD.

For parts e) – g), weekly screen time for teenagers averages 28 hours with an SD of 10 hours. Screen time cannot be negative. Do not assume the normal curve.

e) (5 points) Find the Markov bound and the Chebyshev bound for the percentage of teenagers with 70 or more hours of screen time. Which is better?

f) (5 points) What is the largest possible percentage of teenagers with zero screen time?

g) (4 points) Is it possible that 30% of teenagers have 48 or more hours of screen time? Explain.

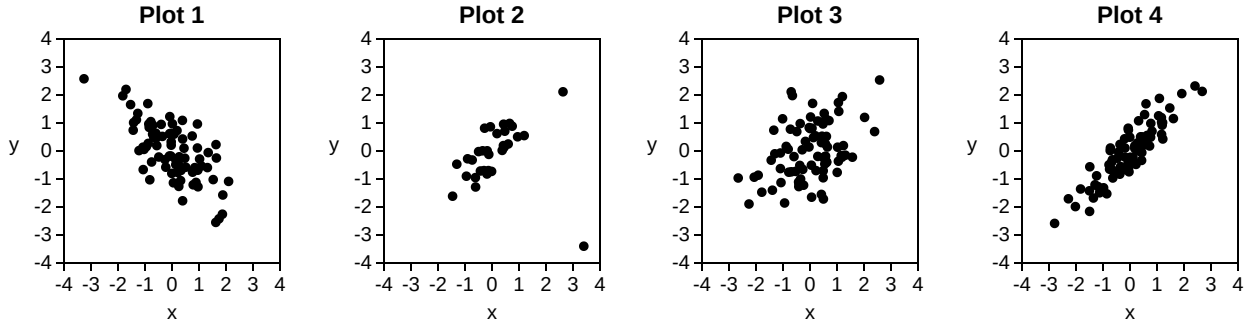
5. (30 points)

a) (8 points) For 50 married couples, husbands' ages have mean 40 and SD 10, wives' ages have mean 38 and SD 9, and the sum of the products of the husband's and wife's ages is $\sum x_i y_i = 79{,}550$. Find the correlation. Would it change if the husbands' ages were measured in months?

b) (10 points) In the table below, (i) find x_3 so that the correlation is −1, or explain why no such value exists; (ii) find x_3 so that the correlation is 0, or explain why no such value exists.

x	2	4	x_3	8
y	10	7	5.5	1

c) (6 points) Match the correlations −0.7, 0.1, 0.4, and 0.9 to the scatter plots below. (No justification needed.)



d) (6 points) (i) Among 8-year-olds the correlation between height and weight is 0.5, and among 12-year-olds it is also 0.5. If we combine the two groups, will the correlation be larger than, smaller than, or about equal to 0.5? Explain. (ii) A large football-shaped data set has correlation 0.8. We keep only the points whose x is above average. What happens to the correlation? Explain.

6. (40 points) In a large study of fathers' heights (x) and their adult sons' heights (y), in inches, the scatter plot is football-shaped. The regression line for predicting a son's height from his father's height is $\hat{y} = 0.5x + 35$. The regression line for predicting a father's height from his son's height is $\hat{x} = 0.5y + 33.5$. The SD of fathers' heights is 3 inches.

a) (8 points) Find the correlation and the average heights of fathers and sons. (Hint: what do you get when you multiply the two slopes? What point lies on both lines?)

b) (6 points) Find the SD of sons' heights and the RMS error for predicting a son's height from his father's height.

c) (8 points) Among fathers who are 74 inches tall, what percentage of sons are taller than their fathers?

d) (6 points) A commentator says: "Sons of tall fathers are shorter than their fathers on average, and sons of short fathers are taller. So over the generations everyone's height is converging, and the spread of heights is shrinking." Comment.

e) (6 points) A father is at the 25th percentile of fathers' heights. Estimate his son's percentile among sons.

f) (6 points) Among sons who are 72 inches tall, estimate the average height of their fathers. Why isn't it 74, even though 74-inch fathers have sons who average 72?

A Normal Table

Height = height of the standard normal curve at z (percent per standard unit). Area = percentage of the area under the normal curve between −z and z.

z	Height	Area	z	Height	Area	z	Height	Area
0.00	39.89	0.00	**1.50**	12.95	86.64	**3.00**	0.443	99.730
0.05	39.84	3.99	**1.55**	12.00	87.89	**3.05**	0.381	99.771
0.10	39.70	7.97	**1.60**	11.09	89.04	**3.10**	0.327	99.806
0.15	39.45	11.92	**1.65**	10.23	90.11	**3.15**	0.279	99.837
0.20	39.10	15.85	**1.70**	9.40	91.09	**3.20**	0.238	99.863
0.25	38.67	19.74	**1.75**	8.63	91.99	**3.25**	0.203	99.885
0.30	38.14	23.58	**1.80**	7.90	92.81	**3.30**	0.172	99.903
0.35	37.52	27.37	**1.85**	7.21	93.57	**3.35**	0.146	99.919
0.40	36.83	31.08	**1.90**	6.56	94.26	**3.40**	0.123	99.933
0.45	36.05	34.73	**1.95**	5.96	94.88	**3.45**	0.104	99.944
0.50	35.21	38.29	**2.00**	5.40	95.45	**3.50**	0.087	99.953
0.55	34.29	41.77	**2.05**	4.88	95.96	**3.55**	0.073	99.961
0.60	33.32	45.15	**2.10**	4.40	96.43	**3.60**	0.061	99.968
0.65	32.30	48.43	**2.15**	3.96	96.84	**3.65**	0.051	99.974
0.70	31.23	51.61	**2.20**	3.55	97.22	**3.70**	0.042	99.978
0.75	30.11	54.67	**2.25**	3.17	97.56	**3.75**	0.035	99.982
0.80	28.97	57.63	**2.30**	2.83	97.86	**3.80**	0.029	99.986
0.85	27.80	60.47	**2.35**	2.52	98.12	**3.85**	0.024	99.988
0.90	26.61	63.19	**2.40**	2.24	98.36	**3.90**	0.020	99.990
0.95	25.41	65.79	**2.45**	1.98	98.57	**3.95**	0.016	99.992
1.00	24.20	68.27	**2.50**	1.75	98.76	**4.00**	0.013	99.9937
1.05	22.99	70.63	**2.55**	1.54	98.92	**4.05**	0.011	99.9949
1.10	21.79	72.87	**2.60**	1.36	99.07	**4.10**	0.009	99.9959
1.15	20.59	74.99	**2.65**	1.19	99.20	**4.15**	0.007	99.9967
1.20	19.42	76.99	**2.70**	1.04	99.31	**4.20**	0.006	99.9973
1.25	18.26	78.87	**2.75**	0.91	99.40	**4.25**	0.005	99.9979
1.30	17.14	80.64	**2.80**	0.79	99.49	**4.30**	0.004	99.9983
1.35	16.04	82.30	**2.85**	0.69	99.56	**4.35**	0.003	99.9986
1.40	14.97	83.85	**2.90**	0.60	99.63	**4.40**	0.002	99.9989
1.45	13.94	85.29	**2.95**	0.51	99.68	**4.45**	0.002	99.9991